Viability Gates: Physical Constraints by (R,N)
Re > 5e+04 (transitional), Tip Speed < 120 m/s (Mach 0.3)

R (m)	N	Configs	Viable	Min Re	Max Tip (m/s	Re OK?	Tip OK?	Viable?
1.5	1	16	16	15000	2	✗	✓	✗
1.5	2	16	16	15000	2	✗	✓	✗
1.5	3	16	16	15000	2	✗	✓	✗
1.5	4	16	16	15000	2	✗	✓	✗
2.0	1	24	24	259546	36	✓	✓	✓
2.0	2	31	31	205740	38	✓	✓	✓
2.0	3	32	32	206470	38	✓	✓	✓
2.0	4	32	32	206283	38	✓	✓	✓
3.0	1	32	32	209820	36	✓	✓	✓
3.0	2	32	32	209868	39	✓	✓	✓
3.0	3	32	32	210163	39	✓	✓	✓
3.0	4	32	32	210163	39	✓	✓	✓

IMPLICATIONS: R=1.5m fails the Re gate at all N — the NACA 0012 operates in the laminar/transitional regime (Re < 5e+04) where CL/CD data is unreliable. R=2.0m passes Re at N≥2 and passes tip speed everywhere. R=3.0m passes all gates at all wind speeds. No configuration exceeds the 120 m/s noise limit — autorotation tip speeds are well below Mach 0.3. The primary viability constraint is minimum Reynolds number, not noise.
Data: BEM v2.1, all wind speeds 6–12 m/s.